

ASSESSMENT TOOL 2 – OUTPUT

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COURSE CODE : CSA1717

COURSE NAME : ARTIFICIAL INTELLIGENCE

QUESTION 1: Constraint Satisfaction Problem (Doctor Shift Scheduling)

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[Step 1] Attempting assignments for variable: D1
-> Testing D1 = Morning: Valid

[Step 2] Attempting assignments for variable: D2
-> Testing D2 = Morning: Violation: Shift already assigned to another doctor
-> Testing D2 = Afternoon: Valid

[Step 3] Attempting assignments for variable: D3
-> Testing D3 = Afternoon: Violation: Shift already assigned to another doctor
-> Testing D3 = Night: Valid

--- FINAL SCHEDULING RESULT ---
D1 -> Morning Shift
D2 -> Afternoon Shift
D3 -> Night Shift
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QUESTION 2: Grid Navigation using Best-First / A* Search

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Start Node: (0, 0), Goal Node: (4, 4)
Obstacles: {(3, 1), (1, 1), (2, 1)}

[Iteration 1] Expanded Node: (0, 0) | g=0, h=8, f=8
-> Generated Neighbor: (1, 0) (g=1, h=7, f=8)
-> Generated Neighbor: (0, 1) (g=1, h=7, f=8)
[Iteration 2] Expanded Node: (0, 1) | g=1, h=7, f=8
-> Generated Neighbor: (0, 2) (g=2, h=6, f=8)
[Iteration 3] Expanded Node: (1, 0) | g=1, h=7, f=8
-> Generated Neighbor: (2, 0) (g=2, h=6, f=8)
[Iteration 4] Expanded Node: (0, 2) | g=2, h=6, f=8
-> Generated Neighbor: (1, 2) (g=3, h=5, f=8)
-> Generated Neighbor: (0, 3) (g=3, h=5, f=8)
[Iteration 5] Expanded Node: (2, 0) | g=2, h=6, f=8
-> Generated Neighbor: (3, 0) (g=3, h=5, f=8)
[Iteration 6] Expanded Node: (0, 3) | g=3, h=5, f=8
-> Generated Neighbor: (1, 3) (g=4, h=4, f=8)
-> Generated Neighbor: (0, 4) (g=4, h=4, f=8)
[Iteration 7] Expanded Node: (1, 2) | g=3, h=5, f=8
-> Generated Neighbor: (2, 2) (g=4, h=4, f=8)
-> Generated Neighbor: (1, 3) (g=4, h=4, f=8)
[Iteration 8] Expanded Node: (3, 0) | g=3, h=5, f=8
-> Generated Neighbor: (4, 0) (g=4, h=4, f=8)
[Iteration 9] Expanded Node: (0, 4) | g=4, h=4, f=8
-> Generated Neighbor: (1, 4) (g=5, h=3, f=8)
[Iteration 10] Expanded Node: (1, 3) | g=4, h=4, f=8
-> Generated Neighbor: (2, 3) (g=5, h=3, f=8)
-> Generated Neighbor: (1, 4) (g=5, h=3, f=8)
[Iteration 11] Expanded Node: (1, 3) | g=4, h=4, f=8
-> Generated Neighbor: (2, 3) (g=5, h=3, f=8)
-> Generated Neighbor: (1, 4) (g=5, h=3, f=8)
[Iteration 12] Expanded Node: (2, 2) | g=4, h=4, f=8
-> Generated Neighbor: (3, 2) (g=5, h=3, f=8)
-> Generated Neighbor: (2, 3) (g=5, h=3, f=8)
[Iteration 13] Expanded Node: (4, 0) | g=4, h=4, f=8
-> Generated Neighbor: (4, 1) (g=5, h=3, f=8)
[Iteration 14] Expanded Node: (1, 4) | g=5, h=3, f=8
-> Generated Neighbor: (2, 4) (g=6, h=2, f=8)
[Iteration 15] Expanded Node: (1, 4) | g=5, h=3, f=8
-> Generated Neighbor: (2, 4) (g=6, h=2, f=8)
[Iteration 16] Expanded Node: (1, 4) | g=5, h=3, f=8
-> Generated Neighbor: (2, 4) (g=6, h=2, f=8)
[Iteration 17] Expanded Node: (2, 3) | g=5, h=3, f=8
-> Generated Neighbor: (3, 3) (g=6, h=2, f=8)
-> Generated Neighbor: (2, 4) (g=6, h=2, f=8)
[Iteration 18] Expanded Node: (2, 3) | g=5, h=3, f=8
-> Generated Neighbor: (3, 3) (g=6, h=2, f=8)
-> Generated Neighbor: (2, 4) (g=6, h=2, f=8)
[Iteration 19] Expanded Node: (2, 3) | g=5, h=3, f=8
-> Generated Neighbor: (3, 3) (g=6, h=2, f=8)
-> Generated Neighbor: (2, 4) (g=6, h=2, f=8)
[Iteration 20] Expanded Node: (3, 2) | g=5, h=3, f=8
-> Generated Neighbor: (4, 2) (g=6, h=2, f=8)
-> Generated Neighbor: (3, 3) (g=6, h=2, f=8)
[Iteration 21] Expanded Node: (4, 1) | g=5, h=3, f=8
-> Generated Neighbor: (4, 2) (g=6, h=2, f=8)
[Iteration 22] Expanded Node: (2, 4) | g=6, h=2, f=8
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
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-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 24] Expanded Node: (2, 4) | g=6, h=2, f=8
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 25] Expanded Node: (2, 4) | g=6, h=2, f=8
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 26] Expanded Node: (2, 4) | g=6, h=2, f=8
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 27] Expanded Node: (2, 4) | g=6, h=2, f=8
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 28] Expanded Node: (3, 3) | g=6, h=2, f=8
-> Generated Neighbor: (4, 3) (g=7, h=1, f=8)
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 29] Expanded Node: (3, 3) | g=6, h=2, f=8
-> Generated Neighbor: (4, 3) (g=7, h=1, f=8)
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 30] Expanded Node: (3, 3) | g=6, h=2, f=8
-> Generated Neighbor: (4, 3) (g=7, h=1, f=8)
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 31] Expanded Node: (3, 3) | g=6, h=2, f=8
-> Generated Neighbor: (4, 3) (g=7, h=1, f=8)
-> Generated Neighbor: (3, 4) (g=7, h=1, f=8)
[Iteration 32] Expanded Node: (4, 2) | g=6, h=2, f=8
-> Generated Neighbor: (4, 3) (g=7, h=1, f=8)
[Iteration 33] Expanded Node: (4, 2) | g=6, h=2, f=8
-> Generated Neighbor: (4, 3) (g=7, h=1, f=8)
[Iteration 34] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 35] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 36] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 37] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 38] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 39] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 40] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 41] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 42] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 43] Expanded Node: (3, 4) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 44] Expanded Node: (4, 3) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 45] Expanded Node: (4, 3) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 46] Expanded Node: (4, 3) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 47] Expanded Node: (4, 3) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 48] Expanded Node: (4, 3) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 49] Expanded Node: (4, 3) | g=7, h=1, f=8
-> Generated Neighbor: (4, 4) (g=8, h=0, f=8)
[Iteration 50] Expanded Node: (4, 4) | g=8, h=0, f=8

--- GOAL REACHED ---
Optimal Path: (0, 0) -> (0, 1) -> (0, 2) -> (0, 3) -> (0, 4) -> (1, 4) -> (2, 4) -> (3, 4) -> (4, 4)
Total Optimal Cost: 8

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QUESTION 3: Autonomous Rescue Robot in Partially Observable Environment

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Start Position: (0, 0), Survivor Position: (3, 3)

[Current Position]: (0, 0) | Cumulative Cost: 0
-> Moving to (0, 1) (Step Cost: 1)

[Current Position]: (0, 1) | Cumulative Cost: 1
-> Moving to (0, 2) (Step Cost: 1)

[Current Position]: (0, 2) | Cumulative Cost: 2
-> Moving to (1, 2) (Step Cost: 1)

[Current Position]: (1, 2) | Cumulative Cost: 3
-> Moving to (1, 3) (Step Cost: 1)

[Current Position]: (1, 3) | Cumulative Cost: 4
-> Moving to (0, 3) (Step Cost: 1)

[Current Position]: (0, 3) | Cumulative Cost: 5
-> Dead end encountered! Backtracking...

[Current Position]: (1, 3) | Cumulative Cost: 5
-> Moving to (2, 3) (Step Cost: 1)

[Current Position]: (2, 3) | Cumulative Cost: 6
-> Moving to (3, 3) (Step Cost: 1)

--- RESCUE OPERATION COMPLETE ---
Path Traversed: (0, 0) -> (0, 1) -> (0, 2) -> (1, 2) -> (1, 3) -> (2, 3) -> (3, 3)
Total Path Cost: 9
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